Social Learning in Non-Stationary Environments

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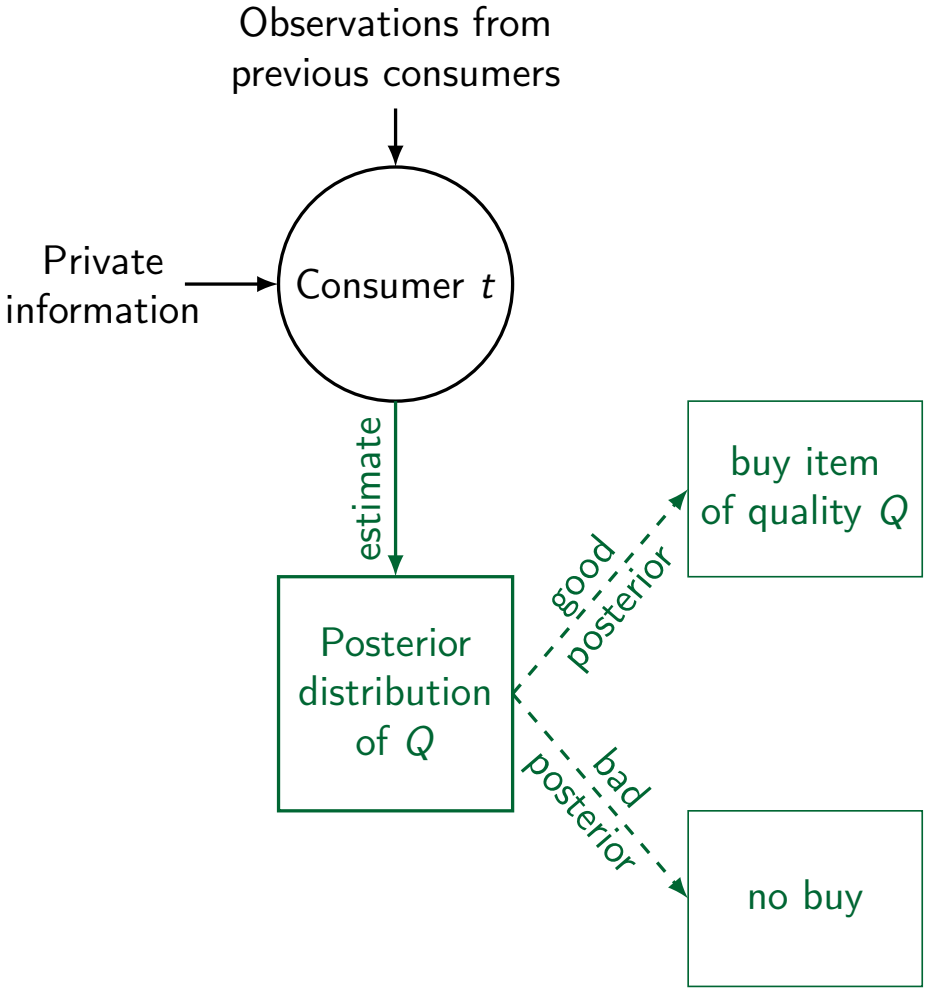
Introduction

Social Learning: learn collectively despite myopic behaviors of individuals

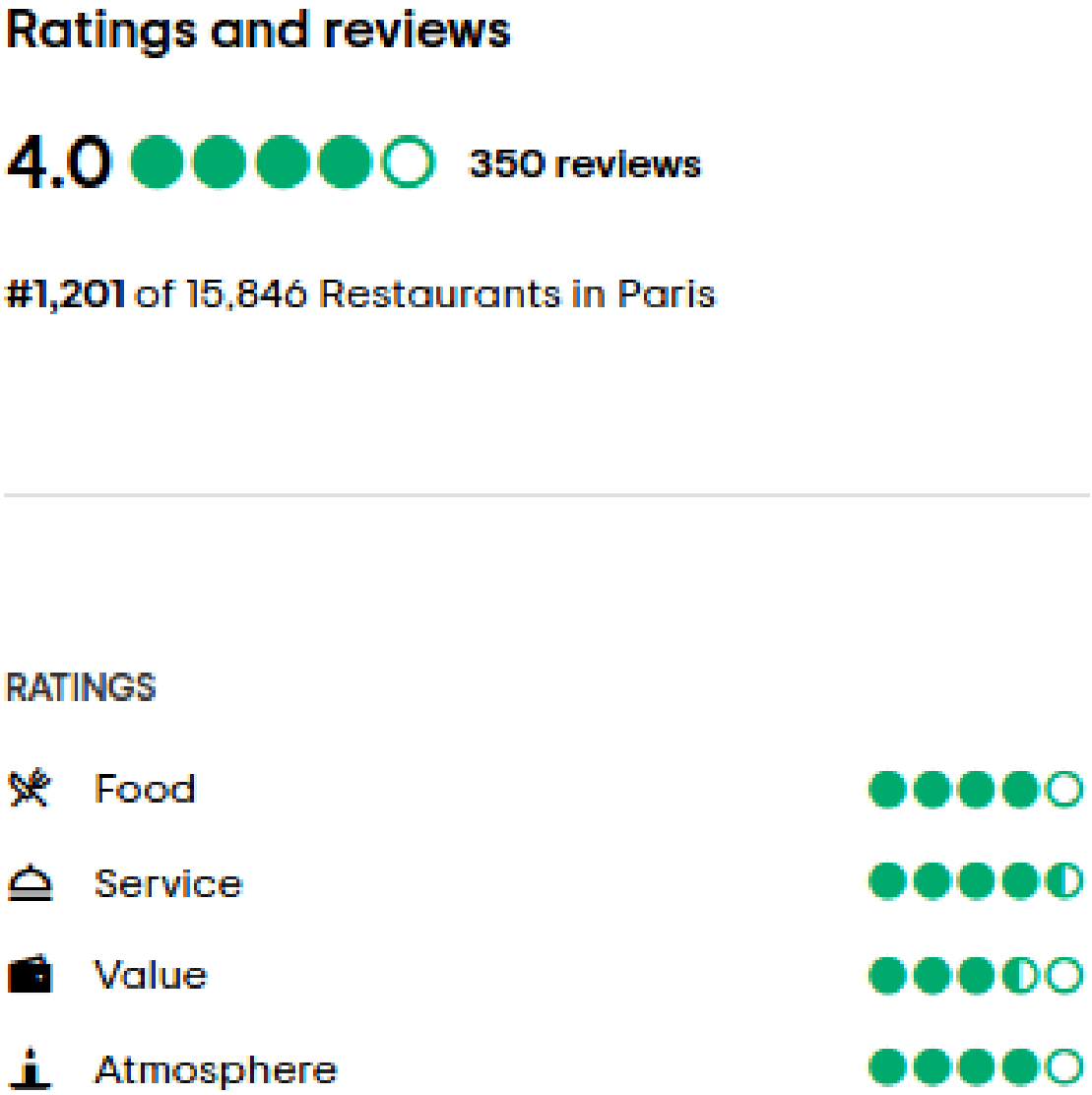
Classical model

- item of quality Q
- consumers arrive sequentially. User t
  - observes decisions of previous consumers
  - receives private signal on quality
  - decides whether to buy or not

Cascades of bad decisions can happen



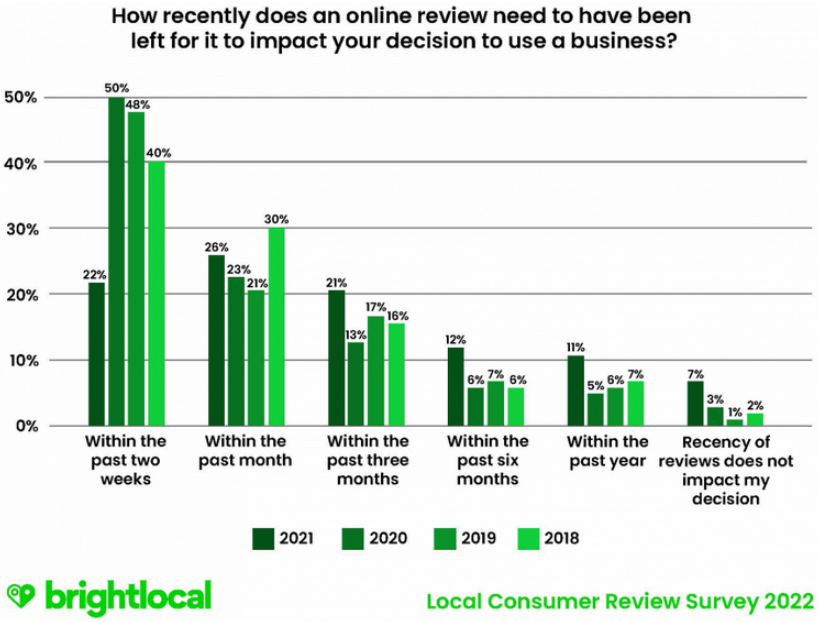
Motivation: learning from reviews



Recent models motivated by online review platforms

- observe experience of previous buyers
- multidimensional quality to estimate
- heterogeneous consumers

Motivation: dynamical environment



- recent reviews have a larger impact on decisions
- possible reason: quality is not fixed but changes over time

Motivation: dynamical environment

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Questions

- does social leaning happen despite individual behaviors?
- if so, what is the rate of learning?
- can a dynamical quality still *be learnt*?

Model

Quality $Q_t \in \{0,1\}^d$

Consumer t has (private) preference $\theta_t \overset{\text{i.i.d.}}{\sim} \Theta$

Utility $u_t = \begin{cases} r(Q_t, \theta_t) \text{ if buy} \\ 0 \text{ otherwise} \end{cases}$

Rational (Bayesian) consumers: buy if and only if $\mathbb{E}[r(Q_t, \theta_t) \mid \mathcal{H}_t, \theta_t] > 0$

$\mathcal{H}_t$ = reviews of previous buyers

After buying, give review

$$Z_t = f(Q_t, \theta_t, \underbrace{\varepsilon_t}_{\text{noise}}) = \left(\underbrace{f^{(i)}(Q_t^{(i)}, \varepsilon_t, \theta_t)}_{\text{review of feature } i} \right)_{i=1,\ldots,d}$$

Difficulties:

- myopic behaviors can hinder learning
- non-i.i.d. observations (buy decisions depend on past reviews)

Model

Assumptions

1. **Purchase guarantee:** r is weakly increasing in each $q^{(i)}$ and $\mathbb{P}_{\theta_t}(r(q, \theta_t) > 0) > 0$ for any q.
2. **Identifiability:** for $q^{(i)} \neq q'^{(i)}$, there exists $z^{(i)}$ such that $\mathbb{P}[Z_t^{(i)} = z^{(i)} | \pi_t = \pi, Q_t^{(i)} = q^{(i)}] \neq \mathbb{P}[Z_t^{(i)} = z^{(i)} | \pi_t = \pi, Q_t^{(i)} = q'^{(i)}]$

1. $\implies$ a fraction of consumers still buy, even when posterior is at lowest
2. $\implies$ different qualities have different review distributions

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Setting example (unidimensional):

- $r(Q, \theta) = Q + \theta - p$
- $f(Q, \theta, \varepsilon) = \mathrm{sign}(Q + \theta + \varepsilon - p)$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

Easy extension to multiple dimensions
Harder reward/feedback models possible (e.g. sparse)

Stationary environment

$Q \sim \pi_0$ and $Q_t = Q$ for all t
Posterior update:

$$\pi_{t+1}(q) \propto \mathbb{P}(Z_t \mid \pi_t, Q_t = q)\pi_t(q)$$

Previous works (unidimensional)
(Ifrach et al., 2019) purchase guarantee $\implies$ social learning happens, i.e.

$$\lim_{t\to\infty} \pi_t(Q) = 1 \text{ with probability } 1$$

(Acemoglu et al., 2017) for $Q' \neq Q$

$$\lim_{t\to\infty} \frac{\log(\pi_t(Q'))}{t} = -\mathrm{KL}\left(\mathbb{P}(Z_t|Q_t=Q,\pi_t=\delta_Q)\middle\|\mathbb{P}(Z_t|Q_t=Q',\pi_t=\delta_Q)\right)$$

Stationary environment

Theorem (Anytime convergence)

For $q' \neq q$

$$\mathbb{E}[\pi_{t+1}(q') \mid Q = q] \leq \frac{e^{-ct}}{\max_i \pi_1^{(i)}(q^{(i)})},$$

where the constant c depends on feedback distribution.

Idea of proof ($d = 1$):
By posterior update,

$$\frac{\log(\pi_{t+1}(q))}{\log(\pi_{t+1}(q'))} = \frac{\log(\pi_1(q))}{\log(\pi_1(q'))} + \sum_{s=1}^{t} \frac{\mathbb{P}(Z_t | Q_t = q, \pi_t)}{\mathbb{P}(Z_t | Q_t = q', \pi_t)}$$

$\square$

$\rightarrow$ anytime analysis necessary for dynamical environment

Stationary environment

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Idea of proof ($d = 1$): Azuma Hoeffding
By posterior update,

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□

$\rightarrow$ anytime analysis necessary for dynamical environment

Dynamical environment

Markov chain model

$$\mathbb{P}\left(Q_{t+1} = q' \mid Q_t = q\right) = P_{q,q'}$$
$$\text{with } P(q,q) \geq 1 - \eta$$

η = probability of quality change

Posterior update mitigates effect from old reviews

$$\pi_{t+1}(q) \propto \sum_{q' \in \mathcal{Q}} P(q,q')\mathbb{P}(Z_t \mid \pi_t, Q_t = q')\pi_t(q')$$

Convergence rate is irrelevant. Consider instead regret

$$R_T = \sum_{t=1}^{T} \mathbb{E}[r(Q_t, \theta_t) - u_t]$$

Minimizing regret $\iff$ maximizing social welfare

If r is 1-Lipschitz, $\quad R_T \leq \sum_{t=1}^{T} \sum_{i=1}^{d} \mathbb{E}[1 - \pi_t(i)(Q_t(i))]$

Dynamical environment

Theorem (Dynamical regret)

$$R_T = \Theta\left(d \log\left(\frac{1}{\eta}\right) \eta T\right).$$

$\eta T \rightarrow$ number of quality blocks

$d \rightarrow$ single block regret in stationary setting

Dynamical environment

Theorem (Dynamical regret)

$$R_T = \Theta\left(d \log\left(\frac{1}{\eta}\right) \eta T\right).$$

$\eta T \rightarrow$ number of quality blocks

$d \log\left(\frac{1}{\eta}\right) \rightarrow$ single block regret in dynamical setting

$\log\left(\frac{1}{\eta}\right) \rightarrow$ time to forget posterior of past block

Additional cost due to dynamical environment $\approx \log\left(\frac{1}{\eta}\right)$

Proof bounds the regret for imperfect consumers (modified transitions $\tilde{P}$)

Naive learners

Question: can imperfect learners still perform social learning?

Naive learners unaware of dynamical structure

$$\pi_{t+1}^{\text{naive}}(q) \propto \mathbb{P}(Z_t \mid \pi_t^{\text{naive}}, Q_t = q)\pi_t^{\text{naive}}(q)$$

Theorem (Naive regret)

$$R_T^{naive} = \Omega(dT)$$

Idea of proof

Time to forget posterior from previous block is too long
$\rightarrow$ huge loss because of inadaptivity ($\sim \frac{1}{\eta}$ per block)

Conclusion

Summary

- anytime analysis of social learning from reviews
- dynamical environment $\leftrightarrow$ signifiance of recency in reviews
- successful social learning for Bayesian consumers in dynamical environment

Type of model	**Utility Loss**
stationary	$\Theta(d)$
dynamical	$\Theta(d \ln(1/\eta)\eta T)$

Perspectives

- can imperfect consumers efficiently learn?
- continuous quality
- seller strategy